

AE-579: Irradiation of [Ru(bpy)3] after irradi of 300 μM Ru(bpy)

Date: 2025-10-08

Tags: O2 [Ru(bpy)3]Cl2*6 H2O AE HTE
Irradiation O2 sensor Power
measurement manual test Autocatalytic
deactivation deactivation

Category: HTE

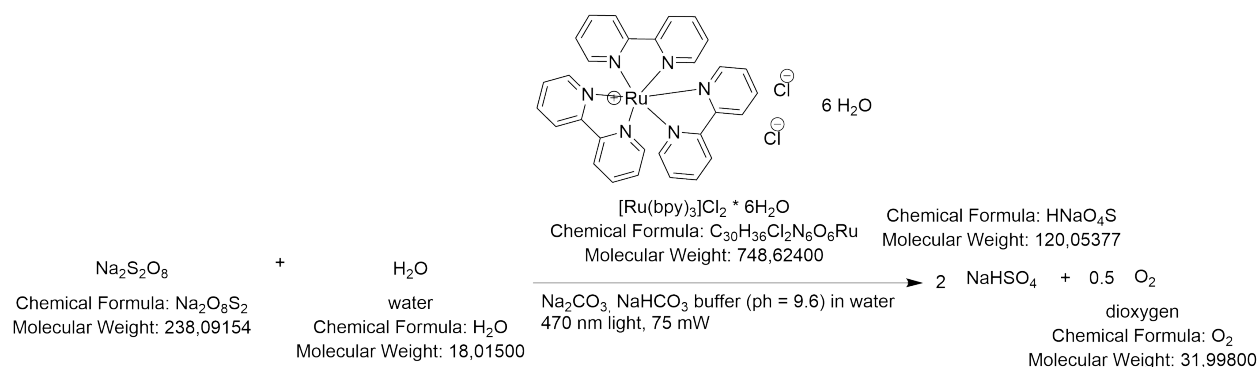
Status: Done

Created by: Alexander Eith

Objective

Repeat AE-566 to test reproducibility with some improvements in experimental procedure

Reaction scheme/sample structure



ChemDraw File (linked): [AE-455.cdxml](#)

Literature/reference experiments

Literature	https://doi.org/10.1021/acscatal.6b02595
Reproduction	/
Similar experiments	HTE - AE-566: Irradiation of [Ru(bpy)3] after irradi of 300 μM Ru(bpy) I

Reagents

Name	CAS Number / Experiment Number	Inventory number	Amount [mmol]	Equivalents	Mass _{exp} [mg]	Used concentration	Molar mass [g/mol]	Density (g/ml)	Volume [ml]	Obtained Concentration in reaction vial [mM]
Na2S2O8	HTE - AE-576: Preparation of stock solutions for the irradiation of [Ru(bpy)3]Cl2 * 6 H2O I (Na2S2O8)	/	/	/	/	60 mM	238.09	/	0.500	6 mM

NaHCO3	HTE - AE-576: Preparation of stock solutions for the irradiation of [Ru(bpy)3]Cl2 * 6 H2O I (NaHCO3)	/	/	/	/	0.9 M	84.006	/	0.381	0.0708 M
Na2CO3	HTE - AE-576: Preparation of stock solutions for the irradiation of [Ru(bpy)3]Cl2 * 6 H2O I (Na2CO3)	/	/	/	/	0.8 M	105.988	/	0.196	0.0292 M
[Ru(bpy)3]Cl2 *6 H2O	HTE - AE-576: Preparation of stock solutions for the irradiation of [Ru(bpy)3]Cl2 * 6 H2O I (Ru)	/	/	/	/	/	/	/	0.75 + 7.5 µL	300 + 10 µM
Milli-Q water	/	/	/	/	/	/	/	/	3.173	/

Irradiation Parameters

Power measured using [Power Meter - 843-R-USB + 919P-020-12](#) in [Equipment - Advanced power measurment setup V1.0](#)

	Name
Used Set-up	Equipment - Advanced irradiation setup V1.0 I
Irradiation setup	Equipment - Irradiation setup, 2

	Light Source Name	Used power source	Wavelength [nm]	Power Setting [mW]	Analogue Setting [-]
First light source	Light Source - UHP LED 470 nm	Power Sources - BLS-13000-1E	470	75	0.28

Used beam combiner [Name or None]	none
Irradiation distance [cm]	6.5
Thermostat temperature [°C]	25
Stirring speed [rpm]	300

Start time [s], relative to start of log	specified in procedure
End time [s], relative to start of log	specified in procedure

O2/H2 sensor equipment

	Equipment	Used Protocol
Used FireSting	Equipment - Firesting Fiber-Optic Oxygen Meter 2 Channel (Firesting 2)	Protocol - Degassing and O2 Measurement with sensor cap with air stable substances
Used O2 Sensor	Equipment - Robust probe for liquid O2 measurment	/
Used H2 Sensor	/	/

Procedure/observations

Date	Time	Step	Observations	Pictures
08.10	09:05	In a 10 mL vial water , NaHCO ₃ , Na ₂ CO ₃ , Ox and Ru where added	AE-579-1 yellow-orange solution	/
	09:10	AE-579-1 was vortexed (Equipment - VWR® VV3, Vortex Mixer, stage 4/6) for apporx. 10 s	/	/
	09:11	From AE-579-1 an aliquote (0.4 mL) was taken	AE-579-300-1	/
	09:25	The power of the light source was adjusted to 75 mW in Equipment - Advanced power measurment setup V1.0 I	/	/
	09:28	In a 5 mL GL14 PTFE Schlenk flask AE-579-1 (2.0 mL) was added using a 1 mL Eppendorf pipette. The flasks was equipped with a Equipment - Robust probe for liquid O2 measurment and placed in the Equipment - Advanced irradiation setup V1.0 I	AE-579-300	/

09:31	The degassing of AE-579-300 was started according to Protocol - Degassing and O2 Measurement with sensor cap with air stable substances	no change	degas 1.jpg
09:38	The oxygen logging was started	/	/
09:48	The degassing was stopped	/	/
09:49	The logging was stopped	/	/
09:49	The oxygen logging was started	/	/
09:55	The irradiation was started after 6:15	/	irrad 1.jpg
10:18	The irradiation was stopped after 29:15	/	/
10:20	To AE-579-300-1 water (0.8 mL) was added using a 1 mL Eppendorf pipette	/	/
10:25	An UV/Vis of AE-579-300-1 was measured (Protocol - UV/vis measurements using Avantes spectrometer, Lab E208, CEEC I)	AE-579-300-1	AE-579-300-1.jpg
10:28	The logging was stopped	darker, more greenish-brownish	/
10:30	0.4 mL of AE-579-300 were transferred into a 5 mL vial	AE-579-300-2	/
	1.5 mL of AE-579-300 were transferred into a 5 mL vial	AE-579-10	/
	To AE-579-10 Ru (7.5 µL) was added	no change	/
	From AE-579-10 0.4 mL were transferred into another vial and 0.8 mL water was added	AE-579-10-1	/
	From AE-579-10 1.0 mL were transferred into the irradiation flasks (which was clean and dried prior using water and kimtech wipes)	AE-579-10	/

		The flask containing AE-579-10 was equipped with a Equipment - Robust probe for liquid O2 measurment and placed in the Equipment - Advanced irradiation setup V1.0 I	/	/
	10:40	The degassing of AE-579-10 was started according to Protocol - Degassing and O2 Measurement with sensor cap with air stable substances	/	/
	10:41	The oxygen logging was started	/	/
	10:42	An UV/Vis of AE-579-300-2 was measured (Protocol - UV/vis measurements using Avantes spectrometer, Lab E208, CEEC I)	AE-579-300-2	AE-579-300-2.jpg
		An UV/Vis of AE-579-10-1 was measured (Protocol - UV/vis measurements using Avantes spectrometer, Lab E208, CEEC I)	AE-579-10-1	/
	10:51	The degassing was stopped	/	/
	10:51	The logging was stopped	/	/
	10:51	The oxygen logging was started	/	setup 2.jpg
	10:59	The irradiation was started after 7:30	/	/
	11:18	The irradiation was stopped after 27:00	/	/
	11:22	The logging was stopped	no significant change	/
		From AE-579-10 0.4 mL were transferred into another vial and 0.8 mL water was added	AE-579-10-2	/
	11:34	An UV/Vis of AE-579-10-2 was measured (Protocol - UV/vis measurements using Avantes spectrometer, Lab E208, CEEC I), for that 0.25 mL was diluted with 1.25 mL Milli-Q water	AE-579-10-2	AE-579-10-2.jpg

Analysis

Used calibration:

- FireSting: [Prep work - AE-401: Liquid-phase calibration of Trace Range Robust Oxygen Probe I](#)

Date	Time	Sample name	Analysis method	Analytical device	Solvent	Raw Data	Python script	Processed Data	Comparative Data	Interpretation
08.10	09:38	AE-579-300-degas	Optical O2 detection	Equipment - Firesting Fiber-Optic Oxygen Meter 2 Channel	H2O	AE-579-300-degas.txt	/	AE-579-300-degas.png	HTE - AE-566: Irradiation of [Ru(bpy)3] after irradiation of 300 µM Ru(bpy) I	worked
08.10	09:49	AE-579-300-irrad	Optical O2 detection	Equipment - Firesting Fiber-Optic Oxygen Meter 2 Channel	H2O	AE-579-300-irrad.txt	/	AE-579-300-irrad.png		Increase till 15 µM
08.10	10:41	AE-579-10-degas	Optical O2 detection	Equipment - Firesting Fiber-Optic Oxygen Meter 2 Channel	H2O	AE-579-10-degas.txt	/	AE-579-10-degas.png		worked
08.10	10:51	AE-579-10-irrad	Optical O2 detection	Equipment - Firesting Fiber-Optic Oxygen Meter 2 Channel	H2O	AE-579-10-irrad.txt	/	AE-579-10-irrad.png		increase till 13 µM, than decrease to 7 µM
08.10	10:25	AE-579-300-1	UV/Vis	Equipment - Avantes UV/vis spectrophotometer	H2O	AE-579-300-1.ABS8 AE-579-300-1_7420287SP.TXT	UV_Vis_multiple_spectra.py	AE-579.png		df = 3 as expected, Ru(II)
08.10	10:42	AE-579-300-2	UV/Vis	Equipment - Avantes UV/vis spectrophotometer	H2O	AE-579-300-2.ABS8 AE-579-300-2_7420287SP.TXT	UV_Vis_multiple_spectra.py	AE-579.png		df = 3 as expected, dimeric / polymeric species
08.10	10:45	AE-579-10-1	UV/Vis	Equipment - Avantes UV/vis spectrophotometer	H2O	AE-579-10-1_7420287SP.TXT AE-579-10-1.ABS8	UV_Vis_multiple_spectra.py	AE-579.png		df = 3 Small increase in orange dimer --> likely shoulder from Ru(II)
08.10	11:34	AE-579-10-2	UV/Vis	Equipment - Avantes UV/vis spectrophotometer	H2O	AE-579-10-2_7420287SP.TXT AE-579-10-2.ABS8	UV_Vis_multiple_spectra.py	AE-579.png		df = 3 significant increase in signal for dimeric/polymeric species

Results

Main outcome as expected (lower O2 rate) than when Ru(II) is directly irradiated.

But O2 formation observed, unlike in [HTE - AE-566: Irradiation of \[Ru\(bpy\)3\] after irradiation of 300 µM Ru\(bpy\) I](#)

Linked experiments

HTE - [AE-547](#): Irradiation of [Ru(bpy)3] with several additions of [Ru(bpy)3]

HTE - [AE-563](#): Irradiation of [Ru(bpy)3] after irradiation of 100 µM Ru(bpy)

HTE - [AE-566](#): Irradiation of [Ru(bpy)3] after irradiation of 300 µM Ru(bpy) I

HTE - [AE-576](#): Preparation of stock solutions for the irradiation of [Ru(bpy)3]Cl₂ * 6 H₂O I

Prep work - [AE-401](#): Liquid-phase calibration of Trace Range Robust Oxygen Probe I

Prep work - [AE-425](#): Liquid-phase calibration of Trace Range Robust Oxygen Probe under static conditions

Linked resources

Equipment - [VWR® VV3, Vortex Mixer](#)

Equipment - [Firesting Fiber-Optic Oxygen Meter 2 Channel \(Firesting 2\)](#)

Equipment - [Robust probe for liquid O₂ measurement](#)

Equipment - [Advanced irradiation setup V1.0 I](#)

Equipment - [Advanced power measurement setup V1.0 I](#)

Equipment - [Irradiation setup 2 \(CEEC II, E004\)](#)

Light Source - [UHP LED 470 nm-1](#)

Power Meter - [843-R-USB + 919P-020-12](#)

Power Sources - [BLS-13000-1E](#)

Protocol - [Degassing and O₂ Measurement with sensor cap with air stable substances](#)

Protocol - [UV/vis measurements using Avantes spectrometer, Lab 208, CEEC I](#)

Attached files

AE-579-10-2.jpg
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setup 2.jpg

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irrad 1.jpg

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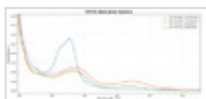
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AE-579.png

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UV_Vis_multiple_spectra.py

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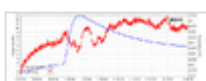
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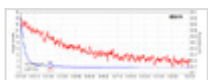


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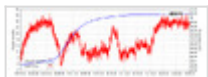


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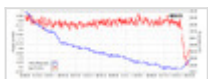
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Link: <https://elab.water-splitting.org/experiments.php?mode=view&id=3125>